Lecture-#3

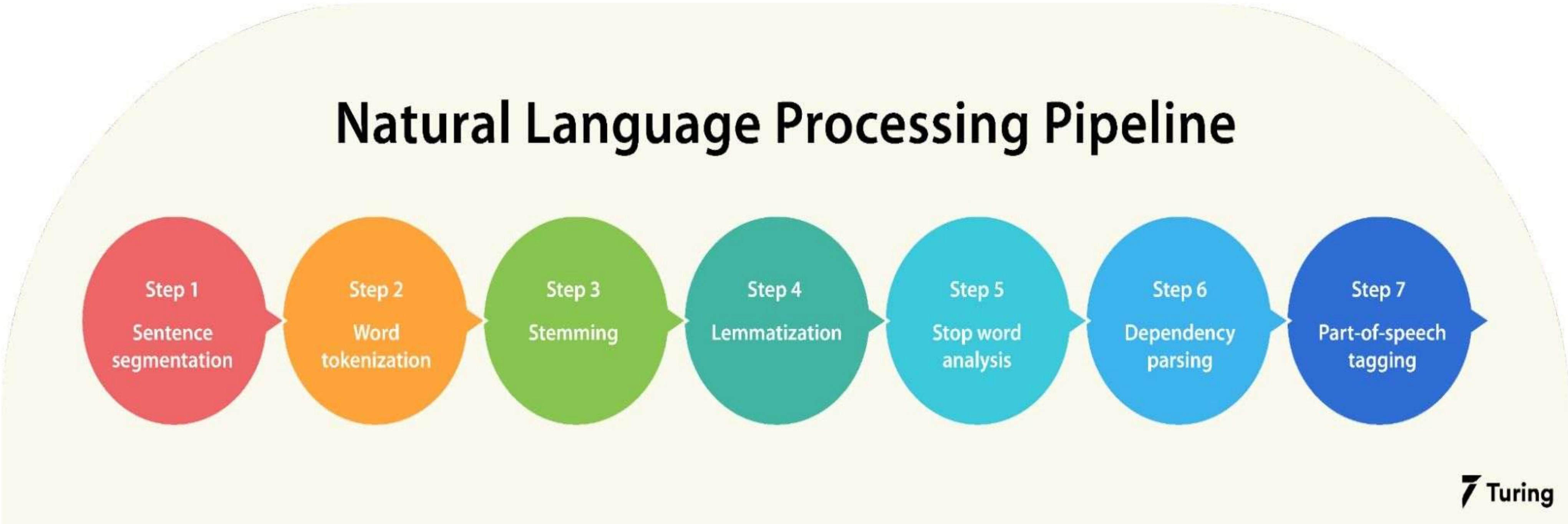
Natural Language Processing

A8706

Introduction

Finding the Structure of Words and Documents

Learning from Experience, Sharing with Passion



Finding the Structure of Words and Documents

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•Human language is a complicated thing.
•We use it to express our thoughts, and through language, we receive information and infer its meaning.
•Trying to understand language all together is not a viable approach.
•Linguists have developed whole disciplines that look at language from different perspectives and at different levels of detail.
•The point of **morphology**, for instance, is to study the variable forms and functions of words,
•The syntax is concerned with the arrangement of words into phrases, clauses, and sentences.
•Word structure constraints due to pronunciation are described by **phonology**,
•The conventions for writing constitute the **orthography** of a language.
•The meaning of a linguistic expression is its semantics, and etymology and lexicology cover especially the evolution of words and explain the semantic, morphological, and other links among them.
•Words are perhaps the most intuitive units of language, yet they are in general tricky to define.
•Knowing how to work with them allows, in particular, the development of **syntactic** and **semantic** abstractions and simplifies other advanced views on language.
•Here, first we explore how to identify words of distinct types in human languages, and how the internal structure of words can be modelled in connection with the grammatical properties and lexical concepts the words should represent.

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- The discovery of word structure is **morphological parsing**.
- In many languages, words are delimited in the orthography by whitespace and punctuation.
- But in many other languages, the writing system leaves it up to the reader to tell words apart or determine their exact phonological forms.

Words and Their Components

- Words are defined in most languages as the smallest linguistic units that can form a complete utterance by themselves.
- The minimal parts of words that deliver aspects of meaning to them are called **morphemes**.

Tokens

- Suppose, for a moment, that words in English are delimited only by whitespace and punctuation (the marks, such as full stop, comma, and brackets)
- Example: Will you read the newspaper? Will you read it? I won’t read it.

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If we confront our assumption with insights from syntax, we notice two here: words ***newspaper*** and ***won't***.

•Being a compound word, ***newspaper*** has an interesting **derivational structure**.

•In writing, *newspaper* and the associated concept is distinguished from the isolated *news* and *paper*.

•For reasons of generality, linguists prefer to analyze *won't* as two syntactic words, or tokens, each of which has its independent role and can be reverted to its normalized form.

•The structure of ***won't*** could be parsed as ***will*** followed by ***not***.

•In English, this kind of tokenization and **normalization** may apply to just a limited set of cases, but in other languages, these phenomena have to be treated in a less trivial manner.

•In Arabic or Hebrew, certain tokens are concatenated in writing with the preceding or the following ones, possibly changing their forms as well.

•The underlying lexical or syntactic units are thereby blurred into one compact string of letters and no longer appear as distinct words.

•Tokens behaving in this way can be found in various languages and are often called **clitics**.

•In the writing systems of Chinese, Japanese, and Thai, whitespace is not used to separate words.

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Lexemes

- By the term word, we often denote not just the one **linguistic form** in the given context but also the **concept behind the form** and the **set of alternative forms** that can express it.
- Such **sets** are called **lexemes or lexical items**, and they constitute the **lexicon** of a language.
- Lexemes can be divided by their behaviour into the lexical categories of verbs, nouns, adjectives, conjunctions, particles, or other parts of speech.
- The citation **form of a lexeme**, by which it is commonly identified, is also called its **lemma**.
- When we convert a word into its other forms, such as turning the **singular *mouse*** into the **plural *mice* or *mouses***, we say we **inflect** the lexeme.
- When we transform a lexeme into another one that is morphologically related, regardless of its lexical category, we say we derive the lexeme: for instance, the nouns ***receiver* and *reception*** are derived from the verb *to **receive***.
- Example: Did you see him? I **didn’t** see him. I didn’t see **anyone**.
- Example presents the problem of tokenization of ***didn’t*** and the investigation of the internal structure of ***anyone***.

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- In the paraphrase ***I saw no one***, the lexeme ***to see*** would be inflected into the form ***saw*** to reflect its grammatical function of expressing **positive past tense**.
- Likewise, ***him*** is the oblique case form of ***he*** or even of a more abstract lexeme representing all personal pronouns.
- In the paraphrase, ***no one*** can be perceived as the minimal word synonymous with ***nobody***.
- The difficulty with the definition of what counts as a word need not pose a problem for the syntactic description if we understand ***no one* as two closely connected tokens treated as one fixed element**.

Morphemes

- Morphological theories differ on whether and how to associate the properties of word forms with their structural components.
- These components are usually called **segments** or **morphs**.
- The morphs that by themselves represent some aspect of the meaning of a word are called **morphemes** of some function.
- Human languages employ a variety of devices by which morphs and morphemes are combined into word forms.

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Morphology

- Morphology is the domain of linguistics that analyses the internal structure of words.
- Morphological analysis – exploring the structure of words
- Words are built up of minimal meaningful elements called morphemes: played = play-ed

cats = cat-s

unfriendly = un-friend-ly

- Two types of morphemes: i Stems: play, cat, friend ii Affixes: -ed, -s, un-, -ly
- Two main types of affixes:

i Prefixes precede the stem: un-

ii Suffixes follow the stem: -ed, -s, un-, -ly

- Stemming = find the stem by stripping off affixes play = play

replayed = re-play-ed

computerized = comput-er-ize-d

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Problems in morphological processing

- **Inflectional morphology**: inflected forms are constructed from base forms and inflectional affixes.
- Inflection relates different forms of the same word

Lemma	Singular	Plural
cat	cat	cats
dog	dog	dogs
knife	knife	knives
sheep	sheep	sheep
mouse	mouse	mice

- **Derivational morphology**: words are constructed from roots (or stems) and derivational affixes:

inter+national = international international+ize = internationalize
internationalize+ation = internationalization

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- The simplest morphological process concatenates morphs one by one, as in dis- agree-ment-s, where <u>agree</u> is a free lexical morpheme and the other elements are bound grammatical morphemes contributing some partial meaning to the whole word.
- In a more complex scheme, morphs can interact with each other, and their forms may become subject to additional phonological and orthographic changes denoted as morphophonemic.
- The alternative forms of a morpheme are termed allomorphs.

Typology

- Morphological typology divides languages into groups by characterizing the prevalent morphological phenomena in those languages.
- It can consider various criteria, and during the history of linguistics, different classifications have been proposed.
- Let us outline the typology that is based on quantitative relations between words, their morphemes, and their features:
- **Isolating**, or **analytic**, languages include no or relatively few words that would comprise more than one morpheme (typical members are Chinese, Vietnamese, and Thai; analytic tendencies are also found in English).

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•**Synthetic** languages can combine more morphemes in one word and are further divided into agglutinative and fusional languages.

Example:

"Domovinosios" (fictional word in a synthetic language inspired by Slavic languages)

- Dom: root meaning "home"
- -ovi: infix denoting "belonging to"
- -nos: suffix denoting plural possession
- -ios: suffix indicating genitive case

- **"Domovinosios"** might be translated as "of the homes" in English

- **Agglutinative** languages have morphemes associated with only a single function at a time (as in Korean, Japanese, Finnish, and Tamil, etc.)

- Example:

"Evlerinizden"

Ev: root meaning "house"

-ler: suffix for plural, meaning "houses"

-iniz: possessive suffix for "your," meaning "your houses"

-den: ablative case suffix, meaning "from"

In English, **"Evlerinizden"** translates to "from your houses."

•**Fusional** languages are defined by their feature-per-morpheme ratio higher than one (as in Arabic, Czech, Latin, Sanskrit, German, etc.).

"Gacchāmi" (गच्छामि)

Gacch: root verb meaning "go"

-ā: suffix indicating the first person singular

-mi: suffix indicating present tense

In this single word, "Gacchāmi" means "I go" or "I am going."

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•**Concatenative** languages linking morphs and morphemes one after another.

"Unhappiness"

Un-: prefix meaning "not" or "opposite of"

Happy: root word meaning "joyful" or "content"

-ness: suffix that turns the adjective "happy" into a noun indicating a state or quality

In this word, "Unhappiness" is formed by concatenating the prefix "Un-", the root "Happy", and the suffix "-ness".

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•**Nonlinear** languages allowing structural components to merge nonsequentially to apply tonal morphemes or change the consonantal or vocalic templates of words.

❖root-and-pattern morphology

•The root **k-t-b** (which generally relates to writing):

•**Kataba** كتب)): "He wrote"

- Root: **k-t-b**
- Pattern: **a-a-a**

•**Kutiba** كُتِبَ)) "It was written"

- Root: **k-t-b**
- Pattern: **u-i-a**

•**Kitaab** كتاب : "Book"

- Root: **k-t-b**
- Pattern: **i-aa**

•In these examples, the consonantal root **k-t-b** remains constant, while different vowel patterns (and sometimes additional affixes) are applied to convey different meanings.

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Morphological Typology

- **Morphological typology** is a way of classifying the languages of the world that groups languages according to their common morphological structures.
- The field organizes languages on the basis of how those languages form words by combining morphemes.
- The morphological typology classifies languages into **two broad classes** of **synthetic languages** and **analytical languages.**
- The **synthetic class** is then further sub classified as either **agglutinative languages** or **fusional languages.**
- Analytic languages contain very little inflection, instead relying on features like word order and auxiliary words to convey meaning.
- Synthetic languages, ones that are not analytic, are divided into two categories: agglutinative and fusional languages.
- Agglutinative languages rely primarily on discrete particles(prefixes, suffixes, and infixes) for inflection, ex: inter+national = international, international+ize = internationalize.
- While fusional languages "fuse" inflectional categories together, often allowing one word ending to contain several categories, such that the original root can be difficult to extract (anybody, newspaper).


Thank you!!!
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